Astronomy/Physics 797 - Galaxies
Syllabus
Spring 2026

Course Title: Galaxies
Instructor: Dr. Gregory Rudnick

Class time and location

Time: T Th 11am - 12:15pm
Place: Malott 2055

Course Hours: 3 credit hours

My Office: 2056B Malott, Phone: 785-864-4099
E-mail: grudnick@ku.edu
Office Hours: I will schedule office hours during the first week of classes. Office hours will be held at the below zoom link or in person in my office at my regular office hours or by appointment.

Class Zoom Link for online participants:
https://kansas.zoom.us/j/89728029550
Passcode: 269043

Slack Workspace invite link: https://join.slack.com/t/astr797galaxy-sp2026/shared_invite/zt-3nk7ewdg3-BenaxeGRBJL9Y_Rvobhhiw

Course Description: This course will cover the concepts that underlie galaxy formation and evolution. We will focus on the observational evidence of how galaxies form and evolve as well as on the theoretical underpinnings of galaxy formation. We will discuss both the individual properties of these other galaxies as well as their ensemble properties. Galaxy evolution is a rich subject with many different facets. I find the synergy of these subjects to be one of the most appealing (for me) parts of this field. A substantial part of the course will be reading and discussing research papers and the course will involve writing a telescope proposal and giving a literature review presentation.

About this class and its learning goals:
The goals of this class are to give you an understanding of:
- observational techniques for studying galaxies;
- the evolution of galaxies as baryon condensates in dark matter haloes with respect to their star formation and stellar population properties;
- the structure and kinematic properties of galaxies, both individually and in terms of galaxy scaling relations;
- the relation of individual properties of galaxies to their population demographics;

- The main theoretical ingredients of galaxy formation;
- how galaxies have transformed over cosmic time.

In addition, you will need to
- Become proficient in calculations related to extragalactic astronomy;
- Learn how to synthesize multiple pieces of information from real research to deduce conclusions from complex and incomplete data;
- Write a persuasive proposal that makes use of archival data to answer a question that is relevant to this class;

Don't deal with challenges on your own

I realize that many of you will be experiencing significant challenges this semester: on your time management, on your home responsibilities, on your finances, and on your physical, emotional, and mental health. While I outline clear expectations for the class in this syllabus, I want to make sure that you can always talk to me should you require an exception. This may be an extension of a due deadline, or an accommodation if you can't make class meetings. Please never hesitate to reach out. I want to help you succeed during this class.

Please get tested for COVID-19 if needed: Testing is a powerful way to avoid transmission to others in class. This may become more important should we experience a resurgence of COVID-19 cases. Information on how to get tested is at https://protect.ku.edu. I have built a course that should enable you to actively participate even if you have to miss class because you are sick, so please get tested if you think you might have been exposed or be sick.

What to do if you are sick: To help ensure the health and safety of our fellow classmates, I do not want you to come to class if you are sick. Please follow the guidelines at https://protect.ku.edu for information about testing.

To assist those who cannot come to class for health reasons, I will be live-streaming all of my lectures via Zoom and will record all lectures. However, coming to class and participating in our many activities will be a critical component to getting a good grade.

If you are too sick to attend class "live", either in person or online, you can take advantage of the recording I will make of each class and will have the opportunity to hand in material late, though you will need to contact me to make those arrangements. Please come to office hours if you have questions about how to catch up.

Mask Policy: To protect all of us, everyone must follow any mask-wearing policies in the classroom as required by the Protect KU Pledge and by University policy. Masks are currently **not** required, but if the university policy changes we will adapt with it. Violations of the mask policy in classrooms are treated as academic misconduct. If you come to class without a mask, I will ask you to put one on. If you do not put on a mask when asked, you will have to leave class. Violations will be reported, and consequences will follow, up to and including suspension from the course.

Diversity, Equity, and Inclusion: It is my intent that students from all diverse backgrounds and perspectives be well served by this course, that students' learning needs be addressed both in and out of class, and that the diversity that students bring to this class be viewed as a resource and benefit.

It is my intent to present materials that are respectful of diversity: gender, sexuality, disability, age, socioeconomic status, ethnicity, race, veteran status, and culture. Your suggestions are encouraged and appreciated.

How to be OK by taking care of yourself . We are all dealing with the stress of everyday life, which can be very different for different people. If you tell me you are having trouble, I'm not going to judge you or think less of you. I hope you will extend me the same grace. Here are some ground rules:

- You never owe me personal information about your health (mental or physical), or anything else. However, you are always welcome to talk to me about things that you might be going through.
- If I can't help you, I will try to direct you to KU resources for assistance.
- If you need extra help, please……just ask! I will listen and will work with you.

Texts and Media:

Required

1. "*Introduction to Galaxy Formation and Evolution: From Primordial Gas to Present-Day Galaxies*", by Cimatti, Fraternali, and Nipoti, Cambridge University Press 2019, ISBN-13 : 978-1107134768. This is a relatively new book that covers galaxy evolution with both a qualitative and quantitative emphasis.

2. A valid e-mail address linked to Canvas that you check at least once a day. I use Canvas to post lecture notes, assignments, solutions, and other relevant course material. You must therefore check Canvas regularly.

3. An account on the class Slack page. On 1/16/26 I sent everyone who was enrolled or who contacted me about auditing.

4. An account on Google Drive. We will use this for some document sharing.

Additional reference materials

The following books are not required, but may serve as useful supplementary texts, especially of your astronomy background is limited. No readings will be assigned in these texts but I can point you to relevant sections should you so desire.

"Galaxy Formation and Evolution", by Hojun Mo, Frank van den Bosch, and Simon White, Cambridge University Press 2010, ISBN 978-0-521-85793-2

Errata can be found at http://people.umass.edu/hjmo/book/errata.pdf

This book is a graduate level textbook on galaxy formation. It has a healthy mix of theory and observations, although it is more biased towards the former.

"Galactic Astronomy", by James Binney and Michael Merrifield, Princeton University Press, 1998, ISBN 0-691-02565-7
A primarily observational overview of galaxies but quite comprehensive. Designed for the graduate level. The only reason that I didn't chose this as the primary text is that it is a little older and misses developments of the last 20 years. But it covers the fundamentals well.

"Galaxies in the Universe: An Introduction" second edition, by Linda S. Sparke and John S. Gallagher III, Cambridge University Press 2007, ISBN-13 978-0-521-67186-6
"*Galaxies in the Universe: An Introduction"* is an undergraduate text that covers galaxies at a more basic level than we will cover in this course.

What you will need to know for this class: This course is designed as a graduate course in galaxy formation. It assumes familiarity with common astrophysical concepts equivalent to that in ASTR 391 as well as in ASTR 691 or 692. This implies some facility with the basics of stellar astronomy. In comparison to ASTR 592 "galactic and extragalactic astronomy", this course will focus on other galaxies as opposed to our own, and will assume a higher proficiency in math appropriate for beginning graduate students, e.g. multi-dimensional integrals and elementary differential equations. I will also have a higher standard for ability to read and digest scientific literature.

A modest ability in programming is preferred as I would like to assign some problems that involve using real data to solve astrophysical problems. If you do not have access to a computer, please contact me as there are computers in the departmental computer lab that you can use.

Accessing articles going back 100 years can be done with the SAO/NASA Astrophysics Data System (ADS; https://ui.adsabs.harvard.edu/) or the arXiv e-print server (http://arxiv.org). Let me know if you need help accessing this.

Relevant textbook chapters and other resources: For additional information on current research I have also posted on Canvas under Course Documents a list of all the Annual Review of Astronomy and Astrophysics papers on galaxies in the last 20 years. These are a good reference for finding the most important papers in a given field. You can access these articles using ADS or arXiv. *This list may not have been posted by the beginning of the class but will be posted soon.*

Course Assignments and Requirements

Lecture format: My lectures will be primarily in Keynote (Mac version of powerpoint). I will post PDF copies of my slides on blackboard after each class, as well as some of the animations that I show (or links to them). Class material will often be posted on Canvas so you should check that frequently (daily) for updates.

Ask questions: I find the material in the class very interesting and I think the class will benefit from an interactive environment. **I encourage you to ask questions whenever you have them** and I anticipate that the unexpected questions and discussions will lead to the most fruitful insights. That being said I will also be actively questioning you during class and trying to encourage structured discussions among you and your peers.

Attendance and Participation policy. The knowledge and skills you will gain in this course highly depend on your participation in in-class learning activities. At the same time, I would like students who are feeling ill to feel comfortable staying home to protect others. I will live-stream all my classes via Zoom and will record all lectures. Students who miss class will be responsible for watching the recorded lecture and keeping up with all work done, including, but not limited to homework, computer exercises, and observing projects. If you have to miss class I want to work with you to make sure that you don't fall behind. However, you will need to take initiative yourself. *In the end, the more you actively participate in class, the more you will get out of it. Students who frequently miss class will have a hard time getting a good grade.*

Astronomy education has many facets - writing and speaking: In addition to traditional homework problem sets, I also plan on giving you several assignments designed to give practically useful experience on presenting your knowledge. You will need to write a telescope proposal and given an oral presentation (see below for details.)

Some of the homework assignments will also contain a writing component and in that case you will be graded on the quality of the writing as well as on its content.

Grades and improving your performance: Given the small class size, it is possible that the grade distribution could look very peculiar. If, at any time, you feel uncertain about your class standing or your absolute grade, **COME AND TALK TO ME ABOUT IT.**

Office Hours and availability: I will poll the class at the beginning of the semester to find the optimal office hours. Invariably there will be people for whom this time doesn't work. Those people may set up an appointment to meet outside of official office hours. **My office hours will be in person and on Zoom. If you want to connect via Zoom during office hours, send me a message on Slack.**

Reading assignments and optimal use of class time: Galaxy evolution involves a synthesis of many different sub-fields of astronomy: stellar evolution, interstellar medium, cosmology, dynamics, astrochemistry, and atomic and molecular spectroscopy, to name a few. While each of these areas are worthy of study in and of themselves, galaxy evolution requires that we combine them to understand complicated phenomena. One of the most productive uses of class time is therefore to tackle these high-level concepts, going back to cover fundamentals when necessary.

To make this work, you need to come to class prepared each day. To that end, reading assignments will be given out often and a due date will be provided. You should read these thoughtfully.

Sometimes not all of the information in a required reading section will be covered in the corresponding lecture. This is because the sections are interwoven in a way that might not exactly correspond to the order in which it is taught. I will try to never get ahead of the reading but don't worry if it takes more than one class for us to address the material covered in the most recent reading.

Homework: There will about 5 homework assignments. To received full credit, assignments must be handed in at the beginning of class on the day that they are due. Homework solutions will be posted on Canvas after all assignments are handed in. I encourage working together on class assignments but the work you hand in must be your own. If there is evidence to the contrary, i.e. answers appear to be copied from a common source, then I reserve the right to lower your grade or initiate academic misconduct proceedings (see below).

All steps and reasoning must be explicitly stated on all problems. Failure to do so will result in a significant reduction in the grade, even if the answer itself is correct. **All assignments must be handed in as PDFs generated using LaTeX.** I will provide more guidance on this throughout the semester, including templates for your to use.

Assignment reflection and regrade

The purpose of assignments is not to penalize you but rather to give you practice at employing different skills. One set of tools to maximize your learning are the detailed solution sets that I will provide for each assignment. My goal this semester is to incentivize you to maximize your use of the solution sets to understand the solutions to each problem. The incentive is a better grade on the assignment and in the course.

For each assignment you will receive a graded version of that assignment back. Each problem will be graded using rubric elements that address the following questions:

1) Did student make an honest attempt at the answer?
2) Did student obtain the correct answer?
3) Did student show sufficient work?
4) Did student explain their work and rationale for each step, including well-commented code for programming problems?

I will distribute the filled out rubric for your assignment, including some small comments if necessary.

Once you receive the assignment and the solution sets, you will have 1 week to fix any issues with your work. Your assignment will be regraded once it is handed in and you have the potential to receive up to 100% on any problem which you sufficiently corrected. You must submit a list of what was changed along with your revised assignment.

The restrictions on this process are as follows:
- You may use my solution sets as much as you would like, though you cannot copy them verbatim.
- You can only attempt problems for which an honest effort was given on the assignment.
- All regrading needs to be handed in within 1 week of when your assignment was returned to you.
- You must follow any original guidelines in the assignment, e.g. using LaTeX, Jupyter Notebooks, GitHub.
- resubmitted assignments in handed in using github need to have a name ending in “revised”
- Each reworked problem must include its own brief reflection statement as to how you used the solution sets or any other resources.
- You must describe what you changed in each of your reworked problems.

Late policy: If you hand in an assignment late it will reduce by 50% the total number of points that you can receive for the assignment, including the regrade. However, any honest attempt to solve the problem will result in a chance to correct it, so I encourage you to hand it in as best as you can rather than handing it in late.

How you should hand in your revised assignment? For every assignment for which a resubmission and regrade is possible I will be making a separate assignment in Canvas that is worth zero points, to which you will upload your revised version. This resubmission assignment will be due two weeks after I distribute the graded original assignment. I will then use that revised version to update your original grade.

The solution sets will be included as an attachment in the graded assignment.

Telescope proposals and review panel: Each of you will be required to write a proposal on a research topic based on the material covered in the class. Each element of the proposal will have different due dates. I am eager to help you write the best proposal you can, so please come talk to me at each stage if you have difficulties.

Students will write **anonymous** comments on each other's proposal. Your grade on this assignment will come partly from my assessment of your assignment and will be partly based on the thoughtfulness and quality of the critique that you give your fellow classmates. The idea is to mimic the kind of persuasive writing that is a necessary element of performing research, and learn how to critically and constructively view other peoples' work.

Computer exercises: A goal of this class is to teach you some basic practical techniques for studying galaxies, which often involves computers. To that end, at different points throughout the semester you may be required to use certain software packages. I will give you plenty of warning as to which packages will be needed. If you have your own computer you can install these on your own and I can try to help a little if I can. Otherwise, we have macintosh laptops that you can borrow for the purposes of this class. These should have the relevant software.

<u>Exams:</u> There will be two non-cumulative midterms in this class and one cumulative final exam. The final exam will be on **Thursday, May 14, 2026 10:30am-1:00pm in 2055 Malott. The midterms will be take home exams on March 5 and April 16.** The midterms will be open notes and open book but you will not be able to consult with any of your classmates.

Policy regarding AI

In most cases, you may use generative artificial intelligence for your work in this class. That approach has limits, though. I explain more below. We will also talk throughout the semester about how that policy is working and whether we need to make any changes. Here are some specifics.

How to use generative AI ethically. I encourage you to use tools like ChatGPT and Bing Chat for things like generating ideas, outlining steps in a project, finding sources, getting feedback on your writing, and overcoming obstacles on papers and projects. That includes getting suggestions for opening paragraphs, and improving transitions and flow. You should explain how you used generative AI in a reflection statement (see below).

What to avoid. Using ChatGPT and similar tools to avoid the intellectual work of your classes is wrong. If you simply copy what a chatbot creates and turn that in as your own, it will be considered academic misconduct. It is the same as copying and pasting from a webpage or having a friend write a paper for you. All of the assignments in this class are intended to help you develop intellectually and professionally. Learning can be challenging, and working through those challenges – and failures – will help you be more successful in this course, in outher courses, and in your profession.

What's the difference? Instructors and students everywhere are trying to negotiate the boundaries of generative AI in learning. Many areas are murky. Here's how I see things: The work you submit in this class has your name on it and should represent your intellectual efforts. Using generative AI for assistance is much like working with a partner. You exchange ideas and offer feedback to each other, with a goal of improving each other's work. If you ask your partner to do the assignment for you because you are tired or sick, you deserve no credit because the assignment with your name on it is a lie. The same holds if you have generative AI do the work. So always ask yourself: What intellectual work have I done? Is this really my work?

Create reflection statements. Each assignment you turn must include a reflection on how you went about the work. That includes the steps you took to find information, the way you organized your information, and the steps you took in creating the assignment. If you used generative AI, explain which tool you used, how you used it, and to what extent. Also explain what you learned from the process. Were AI tools helpful? If so, what approaches do you want to use next time and what approaches should you avoid? Did the use of generative AI make the assignment feel less like your own work? Or were you able to edit and use other strategies to shape the work into your own?

General rules of conduct and student safety:

University of Kansas standards of conduct must be adhered to at all times. I will pursue the maximum punishments in cases where misconduct has occurred. Misconduct includes completing any exam problems with the help of another student or turning in work that is not your own or was copied from someone else, i.e. you can discuss homework with your classmates but the work needs to be your own.

Our department also has an [Aspirational Code of Conduct] that we will be adopting in this class.

If any required class activity conflicts with a mandated religious observance, you should identify yourself to me near the beginning of the semester so other arrangements may be made.

It is paramount that our class be a safe place for learning. If you ever feel unsafe or if you are ever subjected to sexual harassment (or any other kind of harassment) I urge you to contact me immediately. If you do not feel you can deal with me directly, you should discuss it with the Department Chairman (Dr. Feldman) without delay.

Anonymous Comments may be left on the department web page at this link. These comments are read regularly and are acted on to the best of our ability.

Accommodations for disabilities: The Student Access Center (SAC) coordinates academic accommodations and services for all eligible KU students with disabilities. If you have a disability for which you wish to request accommodations and have not contacted SAC, please do so as soon as possible. They are located in 22 Strong Hall and can be reached at 785-864-4064 (V/TTY). Information about their services can be found at www.access.ku.edu. Please contact me privately in regard to your needs in this course.

Other resources: Assistance with your writing can be obtained through the KU writing center (http://www.writing.ku.edu/). More assistance with academic issues, including help with study habits, time management skills, and tutoring can be obtained through the University Academic Support Centers (https://academicsupport.ku.edu/).

Academic Success - Being successful means being healthy, safe, and feeling like you belong

Being in University is stressful and can tax you in ways you haven't before imagined. You should never feel ashamed or shy if you are having problems that make it hard for you to take full advantage of your education at KU. I am always willing to talk to you if you feel that you are having trouble coping or if you need to talk to someone outside the strict bounds of our class.

The University also has extensive resources to help you if you should encounter difficult times. The following headings on the online form of this syllabus contain links to videos about the respective services and I've also included the URLs

- Counseling and Psychological Services video (http://https//caps.ku.edu/)
- Watkins Health Services video (http://studenthealth.ku.edu/)

I strive to make this class an inclusive environment in which everyone has equal opportunities to succeed. As one small part of these efforts I want to make sure that students have access to resources they need to make sure that they feel like they belong at KU. A wonderful resource in this respect is the KU Office of Multicultural Affairs (OMA; https://oma.ku.edu).

Information about resources for religious observances are at https://diversity.ku.edu/religious-observances

Please contact me if you have any issues that impact your ability to be successful at KU!

Course Schedule

What follows is an approximate schedule for our class. Assignment due dates will be posted on Canvas. This schedule is subject to change but is the plan going into the semester.

Dates	Topics	Notes and Other Class events
	First day of class 1/20/26	
1/20, 1/22	Basic processes and cosmological framework	
1/27, 1/29	Cosmological framework	
2/3, 2/5, 2/10, 2/12	Present-day galaxies as a benchmark for evolutionary studies	
2/17, 2/19, 2/24, 2/26	present-day star-forming galaxies	
3/3, 3/5	present-day elliptical galaxies	3/5 midterm #1
3/10	Galaxy environment	
3/12	Dark matter haloes and structure growth	
3/17, 3/19	Spring Break	
3/24	Cont'd - Dark matter haloes and structure growth	
3/26, 3/31, 4/2, 4/7, 4/9, 4/14, 4/16	Ingredients of galaxy formation theory	4/16 midterm #2
4/21, 4/23	The theory of galaxy formation	
4/28, 4/30, 5/5, 5/7	Observing galaxy evolution	

Evaluation Criteria and Grading Scale

Student Survey of Teaching

You will have multiple opportunities to provide feedback on your experience in this course. Suggestions and constructive criticism are encouraged throughout the course and may be particularly valuable early in the semester. To that end, I will use mid-semester surveys and/or reflection assignments to gather input on what is working well and what could be improved. You will also be asked to complete an end-of-semester, online Student Survey of Teaching, which could inform modifications to this course (and other courses that I teach) in the future.

Grading

All grades will be posted on Canvas. You are strongly encouraged to check your scores in Canvas regularly. All grading will be done on an absolute scale with no curve. The breakdown of grades is as follows:

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item	points	Percent of grade
two midterm exams	200	26.67
Cumulative final exam	150	20
Homework (5x50 pts)	250	33.33
Telescope proposal	150	20
Total	**750**	

There will be plus and minus grades in this class. Letter grades will be assigned with the following percentages based on how many points you received out of the total of 975:

Percentage Range	Point range	Letter Grade
90.0-100	675-750	A
86.66-89.99	650-675	B+
83.33-86.65	625-650	B
80.0-83.32	600-625	B-
76.66-79.99	575-600	C+
73.33-76.65	550-575	C

Percentage Range	Point Range	Letter Grade
70-73.33	525-550	C-
66.66-69.99	500-525	D+
63.33-66.65	475-500	D
60.0-63.32	450-475	D-
0-59.99	0-450	F

Incomplete Grades

You may assigned an 'I' (Incomplete) grade if you are unable to complete some portion of the assigned course work because of an unanticipated illness, accident, work-related responsibility, family hardship, or verified learning disability. An Incomplete grade is not intended to give you additional time to complete course assignments or extra credit unless there is indication that the specified circumstances prevented you from completing course assignments on time.

Important Dates

- **Holidays (that means no class):** March 17 and 19 (Spring Break)
- **Other cancelled classes:** April 16th
- **Final exam** will be on Thursday May 14th 10:30am-1pm in 2055 Malott. This may be a takehome exam

Student Resources and University Policies

- Academic Misconduct
  - https://policy.ku.edu/governance/USRR#art2sect7
- Change of Grade
  - https://policy.ku.edu/registrar/grade-change ;
  - https://policy.ku.edu/governance/USRR#art2sect3
- Commercial Note-Taking
  - https://policy.ku.edu/provost/commercial-note-taking
- Diversity and Inclusion
  - https://policy.ku.edu/provost/diversity-inclusion
- Mandatory Reporting (Title IX/Civil Rights)
  - https://policy.ku.edu/civil-rights/mandatory-reporting
- Nondiscrimination, Equal Opportunity, and Affirmative Action
  - https://policy.ku.edu/IOA/nondiscrimination
- Sexual Harassment
  - https://policy.ku.edu/civil-rights/sexual-harassment
- Student Rights and Responsibilities
  - https://policy.ku.edu/student-affairs/student-code
- Racial and Ethnic Harassment Policy
  - https://policy.ku.edu/civil-rights/racial-ethnic-harassment-policy
- Commitment to Integrity and Ethical Conduct
  - https://policy.ku.edu/Chancellor/statement-commitment-integrity-and-ethical-conduct
- KBOR Statement on Free Expression
  - https://publicaffairs.ku.edu/freedom-of-expression
- Weapons, Including Firearms
  - https://policy.ku.edu/university-kansas-policy-weapons-including-firearms-effective-july-1-2017

This syllabus is subject to change

The information in this syllabus is subject to change if circumstances require it, as determined by the instructor. If it is changed I will provide updates both in class and on Canvas.

Library Resources and Services
KU Libraries have world-class collections and scholarly resources for your papers, projects, and coursework – and friendly staff and librarians with the expertise to help you find exactly what you need. Begin your research online, learn about visiting us in person, or Ask A Librarian for help at https://lib.ku.edu/ .